THICKENINGS INDUCE THICKENINGS ON DIAGONALS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 409-426 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 428-451, the complete author proof (22 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 44 of queue_stacks-stacks-more-morphisms.json (pb=928, stmt 409-426, proof 428-451).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 09ZU: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 364–373. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

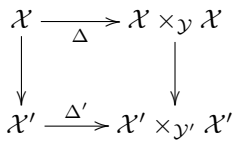
Source reference. Original lemma 0BPE: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 383–387. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CBB: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 451–451. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(f, f') : (\mathcal{X} \subset \mathcal{X}') \to (\mathcal{Y} \subset \mathcal{Y}')$ be a morphism of thickenings of algebraic stacks. Then $\mathcal{X} \times_{\mathcal{Y}} \mathcal{X} \to \mathcal{X}' \times_{\mathcal{Y}'} \mathcal{X}'$ is a thickening and the canonical*

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

diagram



is cartesian.

Proof. Since $\mathcal{X} \to \mathcal{Y}'$ factors through the closed substack $\mathcal{Y}$ we see that $\mathcal{X} \times_{\mathcal{Y}} \mathcal{X} = \mathcal{X} \times_{\mathcal{Y}'} \mathcal{X}$. Hence $\mathcal{X} \times_{\mathcal{Y}} \mathcal{X} \to \mathcal{X}' \times_{\mathcal{Y}'} \mathcal{X}'$ is isomorphic to the composition

$$\mathcal{X} \times_{\mathcal{Y}'} \mathcal{X} \to \mathcal{X} \times_{\mathcal{Y}'} \mathcal{X}' \to \mathcal{X}' \times_{\mathcal{Y}'} \mathcal{X}'$$

both of which are thickenings as base changes of thickenings (Lemma 09ZU). Hence so is the composition (Lemma 0BPE). Since $\mathcal{X} \to \mathcal{X}'$ is a monomorphism, the final statement of the lemma follows from Properties of Stacks, Lemma 0CBB applied to $\mathcal{X} \to \mathcal{X}' \to \mathcal{Y}'$. $\square$